

Docker & Container Commands

1. **Q:** Run a container from an image.
A: `docker run <image>`
2. **Q:** List running containers.
A: `docker ps`
3. **Q:** Stop a container.
A: `docker stop <container_id>`
4. **Q:** Remove a container.
A: `docker rm <container_id>`
5. **Q:** Build an image from a Dockerfile.
A: `docker build -t <name> .`
6. **Q:** Push an image to a registry.
A: `docker push <image>`
7. **Q:** Pull an image from a registry.
A: `docker pull <image>`
8. **Q:** Show container logs.
A: `docker logs <container_id>`
9. **Q:** Inspect container details.
A: `docker inspect <container_id>`
10. **Q:** Execute a command inside a container.
A: `docker exec -it <container_id> /bin/sh`

Kubernetes Basics

11. **Q:** List all pods in default namespace.
A: `kubectl get pods`
12. **Q:** List all namespaces.
A: `kubectl get namespaces`
13. **Q:** Create a resource from a YAML file.
A: `kubectl apply -f <file>.yaml`
14. **Q:** Delete a resource from a YAML file.
A: `kubectl delete -f <file>.yaml`
15. **Q:** Describe a pod.
A: `kubectl describe pod <name>`
16. **Q:** Get logs of a pod.
A: `kubectl logs <pod>`
17. **Q:** Get logs of a specific container in a pod.
A: `kubectl logs <pod> -c <container>`
18. **Q:** Execute a command inside a pod.
A: `kubectl exec -it <pod> -- /bin/sh`
19. **Q:** Scale a deployment to 3 replicas.
A: `kubectl scale deployment <name> --replicas=3`
20. **Q:** Delete a pod.
A: `kubectl delete pod <name>`

⚙️ Kubernetes Advanced

- 21. **Q:** Show cluster nodes.
A: `kubectl get nodes`
- 22. **Q:** Show services in a namespace.
A: `kubectl get svc -n <namespace>`
- 23. **Q:** Show deployments in a namespace.
A: `kubectl get deployments -n <namespace>`
- 24. **Q:** Rollout status of a deployment.
A: `kubectl rollout status deployment <name>`
- 25. **Q:** Undo last deployment rollout.
A: `kubectl rollout undo deployment <name>`
- 26. **Q:** Explain a resource.
A: `kubectl explain <resource>`
- 27. **Q:** Show events in a namespace.
A: `kubectl get events -n <namespace>`
- 28. **Q:** Show resource usage of pods.
A: `kubectl top pod`
- 29. **Q:** Show resource usage of nodes.
A: `kubectl top node`
- 30. **Q:** Apply a label to a pod.
A: `kubectl label pod <name> key=value`

🔒 Security & Config

- 31. **Q:** Create a secret from literal values.
A: `kubectl create secret generic <name> --from-literal=key=value`
- 32. **Q:** Create a ConfigMap from a file.
A: `kubectl create configmap <name> --from-file=<file>`
- 33. **Q:** View secrets in a namespace.
A: `kubectl get secrets`
- 34. **Q:** Decode a secret value.
A: `kubectl get secret <name> -o jsonpath='{.data.key}' | base64 --decode`
- 35. **Q:** View ConfigMaps.
A: `kubectl get configmaps`
- 36. **Q:** Apply a NetworkPolicy.
A: `kubectl apply -f <networkpolicy>.yaml`
- 37. **Q:** Create a ServiceAccount.
A: `kubectl create sa <name>`
- 38. **Q:** Bind a Role to a ServiceAccount.
A: `kubectl create rolebinding <name> --role=<role> -serviceaccount=<namespace>:<sa>`
- 39. **Q:** View RBAC roles.
A: `kubectl get roles`

40. **Q:** View ClusterRoles.

A: `kubectl get clusterroles`

Storage & Stateful

41. **Q:** View PersistentVolumes.

A: `kubectl get pv`

42. **Q:** View PersistentVolumeClaims.

A: `kubectl get pvc`

43. **Q:** Delete a PVC.

A: `kubectl delete pvc <name>`

44. **Q:** Create a StatefulSet.

A: `kubectl apply -f <statefulset>.yaml`

45. **Q:** View StatefulSets.

A: `kubectl get statefulsets`

46. **Q:** View storage classes.

A: `kubectl get storageclass`

47. **Q:** Describe a PV.

A: `kubectl describe pv <name>`

48. **Q:** Describe a PVC.

A: `kubectl describe pvc <name>`

49. **Q:** Delete a PV.

A: `kubectl delete pv <name>`

50. **Q:** Patch a PVC to increase size.

A: `kubectl patch pvc <name> -p`

```
'{"spec":{"resources":{"requests":{"storage":"10Gi"}}}
}'
```

Kubernetes Core

1. **Q:** Create a new namespace.
A: `kubectl create namespace <name>`
2. **Q:** Switch current context to a namespace.
A: `kubectl config set-context --current --namespace=<name>`
3. **Q:** View current context.
A: `kubectl config current-context`
4. **Q:** List all contexts.
A: `kubectl config get-contexts`
5. **Q:** Delete a namespace.
A: `kubectl delete namespace <name>`
6. **Q:** Get YAML output of a pod.
A: `kubectl get pod <name> -o yaml`
7. **Q:** Get JSON output of a service.
A: `kubectl get svc <name> -o json`
8. **Q:** Apply changes interactively.
A: `kubectl edit <resource> <name>`
9. **Q:** Dry-run resource creation.
A: `kubectl apply -f <file>.yaml --dry-run=client`

10. **Q:** Delete all pods in a namespace.
A: `kubectl delete pods --all -n <namespace>`

Deployments & Workloads

11. **Q:** Create a deployment with image nginx.
A: `kubectl create deployment nginx --image=nginx`
12. **Q:** Expose a deployment on port 80.
A: `kubectl expose deployment nginx --port=80`
13. **Q:** Scale deployment to 5 replicas.
A: `kubectl scale deployment nginx --replicas=5`
14. **Q:** Delete a deployment.
A: `kubectl delete deployment <name>`
15. **Q:** Show rollout history of a deployment.
A: `kubectl rollout history deployment <name>`
16. **Q:** Pause a deployment rollout.
A: `kubectl rollout pause deployment <name>`
17. **Q:** Resume a deployment rollout.
A: `kubectl rollout resume deployment <name>`
18. **Q:** Restart a deployment.
A: `kubectl rollout restart deployment <name>`
19. **Q:** Create a job from YAML.
A: `kubectl apply -f <job>.yaml`

20. **Q:** View jobs in a namespace.

A: `kubectl get jobs -n <namespace>`

Security & RBAC

21. **Q:** Create a role with read-only access to pods.

A: `kubectl create role pod-reader --verb=get,list,watch --resource=pods`

22. **Q:** Bind a role to a user.

A: `kubectl create rolebinding <name> --role=pod-reader --user=<username>`

23. **Q:** View rolebindings.

A: `kubectl get rolebindings`

24. **Q:** View clusterrolebindings.

A: `kubectl get clusterrolebindings`

25. **Q:** Delete a rolebinding.

A: `kubectl delete rolebinding <name>`

26. **Q:** Create a TLS secret from certs.

A: `kubectl create secret tls <name> --cert=cert.crt --key=cert.key`

27. **Q:** View service accounts.

A: `kubectl get sa`

28. **Q:** Describe a service account.

A: `kubectl describe sa <name>`

29. **Q:** Annotate a pod.

A: `kubectl annotate pod <name> key=value`

30. **Q:** Remove a label from a pod.

A: `kubectl label pod <name> key-`

Storage & Stateful

31. **Q:** Create a PVC from YAML.

A: `kubectl apply -f <pvc>.yaml`

32. **Q:** Delete a storageclass.

A: `kubectl delete storageclass <name>`

33. **Q:** View volume snapshots.

A: `kubectl get volumesnapshots`

34. **Q:** Create a PV from YAML.

A: `kubectl apply -f <pv>.yaml`

35. **Q:** Patch a deployment to use a PVC.

A: `kubectl patch deployment <name> -p '{"spec":{"template":{"spec":{"volumes":[{"name":"data","persistentVolumeClaim":{"claimName":"mypvc"}}]}}}}'`

36. **Q:** Mount a ConfigMap to a pod.

A: `kubectl apply -f <pod-with-configmap>.yaml`

37. **Q:** Mount a Secret to a pod.

A: `kubectl apply -f <pod-with-secret>.yaml`

38. **Q:** Delete a StatefulSet.

A: `kubectl delete statefulset <name>`

39. **Q:** Scale a StatefulSet.

A: `kubectl scale statefulset <name> --replicas=3`

40. **Q:** View pods of a StatefulSet.

A: `kubectl get pods -l app=<statefulset>`

48. **Q:** Uninstall a Helm release.

A: `helm uninstall <release>`

49. **Q:** Show values of a Helm release.

A: `helm get values <release>`

50. **Q:** Template a Helm chart without installing.

A: `helm template <release> <chart>`

Helm & Ecosystem

41. **Q:** Add a Helm repo.

A: `helm repo add <name> <url>`

42. **Q:** Update Helm repos.

A: `helm repo update`

43. **Q:** Search Helm charts.

A: `helm search repo <chart>`

44. **Q:** Install a Helm chart.

A: `helm install <release> <chart>`

45. **Q:** List Helm releases.

A: `helm list`

46. **Q:** Upgrade a Helm release.

A: `helm upgrade <release> <chart>`

47. **Q:** Rollback a Helm release.

A: `helm rollback <release> <revision>`